NHS

Localisation and adaptation of BTS/NICE/SIGN Asthma Algorithms for use whilst Derbyshire JAPC asthma management guidance is being updated

This resource has been developed as a short-term working document to support clinicians who wish to implement the new national guidance (https://www.nice.org.uk/guidance/ng245) ahead of the updated Derbyshire Joint Area Prescribing Committee Asthma management guidelines which are anticipated to be available in Spring 2025. The expectation is that from a pharmacological management perspective the local guidance will be brought in line with the recommendations from BTS/NICE/SIGN.

Asthma: diagnosis, monitoring and chronic asthma management (BTS/NICE/SIGN) NICE Guideline (NG245) includes summaries of the pharmacological management of asthma in people:

- aged 12 years and over (NG245 Asthma: Algorithm C 27/11/2024)
- aged 5 to 11 years (NG245 Asthma: Algorithm D 27/11/2024) and
- children under 5 years of age (NG245 Asthma: Algorithm E 27/11/2024)

This document is primarily aimed at those aged 12 years and over. The BTS/NICE/SIGN algorithm has been adapted for use in Derbyshire (see page.2) to include information on inhaler licensing and the local formulary and it shows options for specific inhalers for anti-inflammatory reliever (AIR) therapy and maintenance and reliever therapy (MART). The inhaler options included in the locally adapted algorithm are all dry powder inhalers (DPIs). Currently there is no pressurised metered dose inhaler (pMDI) licensed for AIR. A review from 2020 (Climate change in healthcare: Exploring the potential role of inhaler prescribing - Starup-Hansen - 2020 - Pharmacology Research & Perspectives - Wiley Online Library) highlights that DPIs may not be suitable for the following groups:

- frail, elderly patients
- very young patients
- those with muscle weakness

However, for the cohort of patients being considered for this algorithm it is thought that most would be able to use a DPI effectively.

For children aged 5 to 11 years the BTS/NICE/SIGN algorithm for this age group includes the conventional pathway and a MART pathway. Local specialist consensus is that this should be evaluated on a case-by-case basis, with clinicians likely preferring to gain more experience with MART in children over 12 years old before applying it to younger age groups. When more experienced in discussing MART and the application of MART action plans in older children (and adults) clinicians may feel more competent to consider MART use in children aged 5-11 years. The specialist teams are gaining insight into the age range that may be considered able to use a DPI effectively, however they recommend that such patients have the opportunity to demonstrate use and practice on a trainer device. It would also be important to determine how each child and their support network manages their symptoms, considering whether the child requests their current short-acting beta-2 agonist (SABA) independently, while being aware of their triggers and what usually leads them to use an inhaler for symptom relief.

Algorithm C: Pharmacological management of asthma in people aged 12 years and over

BTS, NICE and SIGN guideline on asthma - Adapted for use in DDICB

Take into account and try to address the possible reasons for uncontrolled asthma before starting or adjusting medicines for asthma. For example: alternative diagnoses or comorbidities; suboptimal adherence; suboptimal inhaler technique; active or passive smoking (including e-cigarettes); psychosocial factors; seasonal factors; environmental factors (such as air pollution and indoor mould exposure)

Symptom relief

MART

Maintenance therapy

When changing from low- or moderate-dose ICS (or ICS/LABA combination inhaler) plus supplementary therapy to MART, consider whether to stop or continue the supplementary therapy based on the degree of benefit achieved when first introduced

Existing diagnosis of asthma on the treatment pathway recommended by previous NICE and BTS/SIGN guidelines

SABA only → Offer low-dose ICS/formoterol combination inhaler to be taken as needed (AIR therapy)

With a SABA

If asthma is uncontrolled on:
- Low-dose ICS
- Low-dose ICS/LABA
- Low-dose ICS + LTRA, or
- Low-dose ICS/LABA + LTRA

→ Consider → Low-dose MART

With a SABA

If asthma is uncontrolled on:
- Moderate-dose ICS
- Moderate-dose ICS/LABA
- Moderate-dose ICS + LTRA and/or LAMA
- Moderate-dose ICS/LABA + LTRA and/or LAMA

→ Consider → Moderate-dose MART

If asthma is uncontrolled on high-dose ICS → Refer people to a specialist in asthma care

Newly diagnosed asthma in people aged 12 and over

→ Offer low-dose ICS/formoterol combination inhaler to be taken as needed (AIR therapy)

If asthma is uncontrolled, **offer** → Low-dose MART

If asthma is uncontrolled, **offer** → Moderate-dose MART

If asthma is uncontrolled, despite good adherence → Check FeNO level, if available, and blood eosinophil count

If either is raised → Refer people to a specialist in asthma care

If neither is raised → Consider a trial of either LTRA or LAMA used in addition to moderate-dose MART for 8 to 12 weeks unless there are side effects. At the end of the trial:
- if asthma is controlled, continue the treatment
- if control has improved but is still inadequate, continue the treatment and start a trial of the other medicine (LTRA or LAMA)
- if control has not improved, stop the LTRA or LAMA and start a trial of the alternative medicine (LTRA or LAMA)

If asthma is uncontrolled → Refer people to a specialist in asthma care

If highly symptomatic or there are severe exacerbations, offer low-dose MART → Low-dose MART

If asthma is controlled, consider stepping down

Uncontrolled asthma: Any exacerbation requiring oral corticosteroids **or** frequent regular symptoms (such as using reliever inhaler 3 or more days a week or night-time waking 1 or more times a week)

ICS, inhaled corticosteroid; **LABA**, long-acting beta$_2$ agonist; **LAMA**, long-acting muscarinic receptor antagonist; **LTRA**, leukotriene receptor antagonist; **MART**, maintenance and reliever therapy (using ICS/formoterol combination inhalers); **SABA**, short-acting beta$_2$ agonist.

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Do not prescribe a SABA inhaler for patients using AIR or MART regimens

Licensed inhaler options from DDICB formulary

AIR

WockAIR 160/4.5 DPI

Fobumix 160/4.5 DPI

Duoresp Spiromax 160/4.5 DPI

Symbicort Turbohaler 200/6 DPI

Low-dose MART*

WockAIR 160/4.5 DPI

Fobumix Easyhaler 160/4.5 DPI

Duoresp Spiromax 160/4.5 DPI

Symbicort Turbohaler 200/6 DPI

Fostair Nexthaler 100/6 DPI

Moderate-dose MART*

WockAIR 160/4.5 DPI

Fobumix Easyhaler 160/4.5 DPI

Duoresp Spiromax 160/4.5 DPI

Symbicort Turbohaler 200/6 DPI

*See Pages 3-4 for MART doses

Table 1: Licensed MART Dosing

	Low dose maintenance dose for MART	Moderate dose maintenance dose for MART	Additional doses
WockAir (budesonide/formoterol) **160/4.5**	2 inhalations per day, given either as one inhalation in the morning and evening or as 2 inhalations in either the morning or evening.	2 inhalations twice daily	Patients should take 1 additional inhalation as needed in response to symptoms. If symptoms persist after a few minutes, an additional inhalation should be taken. Not more than 6 inhalations should be taken on any single occasion. A total daily dose of more than 8 inhalations is not normally needed; however, a total daily dose of up to 12 inhalations could be used for a limited period. Patients using more than 8 inhalations daily should be strongly recommended to seek medical advice.
Fobumix Easyhaler (budesonide/formoterol) **160/4.5**	2 inhalations per day, given either as one inhalation in the morning and evening or as 2 inhalations in either the morning or evening.	2 inhalations twice daily	Patients should take 1 additional inhalation as needed in response to symptoms. If symptoms persist after a few minutes, an additional inhalation should be taken. Not more than 6 inhalations should be taken on any single occasion. A total daily dose of more than 8 inhalations is not normally needed; however, a total daily dose of up to 12 inhalations could be used for a limited period. Patients using more than 8 inhalations daily should be strongly recommended to seek medical advice.
Duoresp Spiromax (budesonide/formoterol) **160/4.5**	2 inhalations per day, given either as one inhalation in the morning and evening or as 2 inhalations in either the morning or evening.	2 inhalations twice daily	Patients should take 1 additional inhalation as needed in response to symptoms. If symptoms persist after a few minutes, an additional inhalation should be taken. Not more than 6 inhalations should be taken on any single occasion. A total daily dose of more than 8 inhalations is not normally needed; however, a total daily dose of up to 12 inhalations could be used for a limited period. Patients using more than 8 inhalations daily should be strongly recommended to seek medical advice.
Symbicort Turbohaler (budesonide/formoterol) **200/6**	2 inhalations per day, given either as one inhalation in the morning and evening or as 2 inhalations in either the morning or evening.	2 inhalations twice daily	Patients should take 1 additional inhalation as needed in response to symptoms. If symptoms persist after a few minutes, an additional inhalation should be taken. Not more than 6 inhalations should be taken on any single occasion.

			A total daily dose of more than 8 inhalations is not normally needed; however, a total daily dose of up to 12 inhalations could be used for a limited period. Patients using more than 8 inhalations daily should be strongly recommended to seek medical advice.
Fostair Nexthaler (beclometasone extra-fine particle /formoterol) **100/6**	One puff twice a day	Not within product licence**	Patients should take 1 additional inhalation as needed in response to symptoms. If symptoms persist after a few minutes, an additional inhalation should be taken. The maximum daily dose is 8 inhalations.
Bibecfo/Luforbec/Proxor (beclometasone extra-fine particle /formoterol) **100/6 pMDI** (MDI only to be used where DPI assessed to be unsuitable)	One puff twice a day	Not within product licence**	Patients should take 1 additional inhalation as needed in response to symptoms. If symptoms persist after a few minutes, an additional inhalation should be taken. The maximum daily dose is 8 inhalations.

**Any decision to use would be off label but after individual patient review may be considered appropriate rather than changing device/drug combination

NOTE Symbicort 100/6 and Fobumix 80/4.5 are also licensed for MART but the dose within the licence is equivalent to a very low dose maintenance ICS dose so does not fit directly into the BTS/NICE/SIGN algorithm for pharmacological management of asthma in people aged 12 and over.

	Very low dose maintenance dose for MART	**Additional doses**
Fobumix Easyhaler (budesonide/formoterol) **80/4.5**	2 inhalations per day, given either as one inhalation in the morning and evening or as 2 inhalations in either the morning or evening.	Patients should take 1 additional inhalation as needed in response to symptoms. If symptoms persist after a few minutes, an additional inhalation should be taken. Not more than 6 inhalations should be taken on any single occasion. A total daily dose of more than 8 inhalations is not normally needed; however, a total daily dose of up to 12 inhalations could be used for a limited period. Patients using more than 8 inhalations daily should be strongly recommended to seek medical advice.
Symbicort Turbohaler (budesonide/formoterol) **100/6**	2 inhalations per day, given either as one inhalation in the morning and evening or as 2 inhalations in either the morning or evening.	Patients should take 1 additional inhalation as needed in response to symptoms. If symptoms persist after a few minutes, an additional inhalation should be taken. Not more than 6 inhalations should be taken on any single occasion. A total daily dose of more than 8 inhalations is not normally needed; however, a total daily dose of up to 12 inhalations could be used for a limited period. Patients using more than 8 inhalations daily should be strongly recommended to seek medical advice.

Table 2: Formoterol combination inhaler licensing information

Inhaler	AIR	Licensed age for AIR	MART	Licensed age for MART	Maintenance therapy	Licensed age for maintenance therapy	Doses per device	Cost per device (correct as of January 2025)	Dose counter? (information taken from rightbreathe.com)	Shelf-life
Budesonide + Formoterol										
Fobumix 80/4.5		N/A	✓	12 and over	✓	6 and over	120	£21.50	✓	4 months after opening foil wrapping
Fobumix 160/4.5	✓	N/A	✓	12 and over	✓	12 and over	120	£21.50	✓	
Fobumix 320/9		N/A		N/A	✓	12 and over	60	£21.50	✓	
WockAIR 160/4.5	✓	12 and over	✓	12 and over	✓	12 and over	120 (2x60)	£19.00	✓	2 years
WockAIR 320/9		N/A		N/A	✓	12 and over	60	£19.00	✓	
Duoresp Spiromax 160/4.5	✓	12 and over	✓	12 and over	✓	12 and over	120	£27.97	✓	12 months after opening foil wrap
Duoresp Spiromax 320/9		N/A		N/A	✓	12 and over	60	£27.97	✓	
Symbicort Turbohaler 100/6		N/A	✓	6 and over	✓	6 and over	120	£28.00	✓	3 years
Symbicort Turbohaler 200/6	✓	12 and over	✓	12 and over	✓	12 and over	120	£28.00	✓	
Symbicort Turbohaler 400/12		N/A		N/A	✓	12 and over	60	£28.00	✓	
Symbicort pMDI 100/3	Non-formulary. Note 100/3, not 100/6 as per turbohaler device. Licensed in asthma for maintenance therapy and MART in aged 12 and over. Symbicort pMDI contains propellant HFA227ea which has a significantly higher carbon footprint than other propellants and so should be avoided where possible.						120	£14.00	✓	3 months after first opening
Symbicort pMDI 200/6	LICENSED FOR COPD ONLY - Symbicort pMDI contains propellant HFA227ea which has a significantly higher carbon footprint than other propellants and so should be avoided where possible.						120	£28.00	✓	
Beclometasone extra-fine particle + Formoterol										
Bibecfo/Luforbec/Proxor 100/6 pMDI		N/A	✓	18 and over	✓	18 and over	120	£13.98 (Proxor £9.90)	✓	3 months from the date you get inhaler from pharmacy
Bibecfo/Luforbec/Proxor 200/6 pMDI		N/A		N/A	✓	18 and over	120	£13.98 (Proxor £9.90)	✓	
Fostair 100/6 and 200/6 pMDI	Non-formulary - but to note licensing is the same as Bibecfo/Luforbec/Proxor						120	£29.32	✓	
Fostair Nexthaler 100/6		N/A	✓	18 and over	✓	18 and over	120	£29.32	✓	

Fostair Nexthaler 200/6		N/A		N/A	✓	18 and over	120	£29.32	✓	6 months after opening the pouch

Document control

Amendment	Date
Symbicort 100/6 Turbohaler licensing information updated to state MART now licensed for age 6 years and over	October 2025
Addition of advice not to prescribe SABA with AIR/MART in Algorithm C WockAIR and Fobumix font made bold in Algorithm C inhaler options list to indicate 1st line choices Added extra-fine particle beclometasone wording to Fostair, Luforbec, Proxor and Bibecfo inhalers	March 2026